

Lab 4 – Dimensionality Reduction and MLP

1. Give some intuitive reason why unit hypersphere volume decreases (rather than increases) as # of dimension exceeds 5.
2. Dimensionality Reduction is very important, especially, considering using Machine Learning in Big Data applications. Below are the 2 papers you need to read. These will help better understand what we covered in the class. These would also help you in your project as well as when you will join the industry. The first paper is relative easier than the 2nd one which has more math.
 - a. <https://ieeexplore.ieee.org/document/9036908>
Analysis of Dimensionality Reduction Techniques on Big Data
 - b. <http://jmlr.org/papers/volume16/cunningham15a/cunningham15a.pdf>
**Linear Dimensionality Reduction:
Survey, Insights, and Generalizations**

Just write a short summary of the **first paper based on your own understanding.**

This paper investigates how dimensionality reduction techniques affect the training performance of four popular machine learning algorithms Decision Tree, Naive Bayes, Random Forest, and Support Vector Machine. Evaluate Principal Component Analysis (PCA) and Linear Discriminant Analysis (LDA) across three distinct datasets. Experimental results demonstrate that PCA generally performs better than LDA by preserving important feature relationships instead of aggressively collapsing dimensions. Importantly, the paper finds that PCA combined with Random Forests or SVM yields superior results for large, high-dimensional data sets such as IDS, while applying dimensionality reduction to smaller, lower-dimensional data sets actually degrades model performance.

3. **Write any key points that you have learned from the paper that you can possibly use in your project.** In case you are not addressing any DR for your project now, write how DR can affect your project in future when you may have large data with many dimensions.
- Model accuracy and metrics are improved: applying PCA actually yields higher accuracy, sensitivity, and specificity metrics compared to models trained on the original, uncorrected feature set.
 - Significantly reduces computational burden: Applying techniques like PCA reduces space and time complexity, making training faster and less resource-intensive without sacrificing performance.
 - Optimize with specific Classifiers: Pair DR with powerful algorithms like Random Forests (RF) or Support Vector Machines (SVM) to achieve optimal classification performance and stability.